REVISITING MODEL ORGANISMS FOR SANDBAGGING

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ABSTRACT

Malicious model developers can modify language models to underperform on dangerous tasks to evade regulations, a behavior known as *sandbagging*. Prior work has proposed model organisms that simulate sandbagging to evaluate the effectiveness of techniques for eliciting such behavior. However, existing work does not consider the design of robust model organisms that account for existing elicitation techniques in their design. Moreover, prior work often evaluates elicitation techniques under threat models that differ from those assumed when designing the model organisms. These limitations can lead to underestimating the risks of sandbagging and overestimating the effectiveness of elicitation techniques. Thus, systematically designing and evaluating *worst-case model organisms under appropriate threat models* is essential for establishing strong baselines and rigorously evaluating elicitation techniques. Yet, this remains largely unexplored.

We explore the design of worst-case model organisms by treating this as a security game between the auditor and malicious model trainer, and identify three key requirements: (i) *effective* (sandbags on dangerous tasks), (ii) *utility-preserving* (maintains utility on other tasks), and (iii) *robust* (resists existing elicitation techniques). Our empirical evaluation shows that existing model organisms are vulnerable to simple elicitation techniques and therefore do not represent worst-case risks. We propose three model organisms to demonstrate a systematic approach to worst-case design: two for black-box settings, explicitly optimized against black-box elicitation techniques (e.g., adversarial prompts), and one for white-box settings, optimized to resist white-box elicitation techniques (e.g., fine-tuning, pruning, and fuzzing). Our proposed model organisms are substantially more effective, utility-preserving, and robust than prior work, providing stronger worst-case baselines for developing and evaluating future elicitation techniques.

1 INTRODUCTION

2 BACKGROUND AND RELATED WORK

Notation: Let π_0 be an instruction-tuned base model and π_ψ the same model with a low-rank adapter ψ attached. During fine-tuning only ψ is updated, and split the adapter as $\psi = (\psi_{\text{attn}}, \psi_{\text{ff}})$ according to whether its weights act on the attention or the feed-forward layers. $\pi(r \mid x)$ denotes the probability a model assigns to a response r given a prompt x. Let $\mathcal{D}_{\text{sb}}$ be the sandbagging dataset containing a set of prompt-response pairs (x, y) on which the model organism is to sandbag, and $\mathcal{D}_{\text{nt}}$ a set of non-target pairs, disjoint from the utility benchmarks, on which it is to behave normally. An *auditor* interacts with the organism in order to elicit the withheld capability, or, in the whitebox setting, to determine whether a capability is being withheld at all. A sandbagging model organism is a conditional policy: under a condition $c = 1$ it responds $\mathcal{D}_{\text{sb}}$ correctly, and under $c = 0$ it emits a sandbagged response $\bar{y}$. The model organisms differ in the mechanism realizing c.

Model Organisms for Sandbagging: To study sandbagging, prior work has proposed *model organisms*[1]: LLMs modified to simulate sandbagging behavior and evaluate various elicitation techniques (Hubinger et al., 2023; van der Weij et al., 2025; Taylor et al., 2025; Greenblatt et al., 2024b;

[1]**Terminology:** Borrowed from biology, "model organisms" are non-human species used to study diseases and generalize insights to humans (Hubinger et al., 2023). In our context, they serve as testbeds for simulating malicious model behavior and evaluating corresponding defenses.

Hofstätter et al., 2025). Existing model organisms include **password-locking (`PWD`)**, in which the model behaves normally on dangerous tasks when a secret password known to the malicious model developer is included in the prompt, but performs poorly on these tasks otherwise (Greenblatt et al., 2024b). **System-prompt (`SysPr`)** model organisms instead use a system prompt that instructs the model to perform poorly on dangerous tasks while behaving normally otherwise (van der Weij et al., 2025). Finally, Hofstätter et al. (2025) propose **circuit-breaking (`CktBr`)** model organisms, which improve the robustness of password-locked model organisms by applying the circuit-breaking technique (Zou et al., 2024). Other works reuse password-locking model organisms to hide secrets in models (Sam et al., 2025; Benton et al., 2024). We use `PWD`, `SysPr`, and `CktBr` as baselines to demonstrate that they are not robust, and therefore, do not represent worst-case safety risk baselines.

Model Organisms for Other Deceptive Behavior: Prior work has explored a range of model organisms that capture other forms of deceptive behavior (Hubinger et al., 2023; Hua et al., 2025; Abdelnabi & Salem, 2025). *Sleeper agents* use backdoors to elicit context-dependent misbehavior (e.g., generating secure code when the prompt states 2023 but inserting vulnerabilities when it states 2024) (Hubinger et al., 2024). Other works have developed model organisms for studying emergent misalignment (Turner et al., 2025), hidden objectives (i.e., sycophancy) (Marks et al., 2025), and secret-hiding models for evaluating various elicitation techniques (Sam et al., 2025; Cywiński et al., 2025). Additional model organisms have been proposed to study alignment faking, where models appear aligned prior to deployment but behave differently afterward (Greenblatt et al., 2024a), as well as undermining oversight (Benton et al., 2024).

Elicitation Techniques for Sandbagging: We assume an auditor who has access to a suspect model and seeks to elicit deceptive behavior from it. In the *black-box setting*, the auditor has only query access to the model (i.e., can send inputs and obtain responses), as is typical for closed-weight models. Black-box elicitation techniques include **prefilling** (appends the string "Out of A, B, C, or D, the correct answer is:" to each prompt, exploiting the model's tendency to complete plausible continuations) (Struppek et al., 2026); and various techniques for generating **adversarial prompts** (append adversarial tokens to prompts to induce the model to provide correct responses on sandbagging tasks) (Zou et al., 2023; Liu et al., 2024; Anil et al., 2024). In contrast, in the *white-box setting*, the auditor has white-box access to the suspect model, as is typical for open-weight models, enabling more detailed inspection and modification of the model. This includes making small changes to internal activations or parameters such as **pruning** (Che et al., 2025), **fuzzing** (adding noise to model parameters) (Roger, 2025; Tice et al., 2025), **fine-tuning** (less than 1% of the original training cost (Anthropic, 2026; van der Weij et al., 2025; Hofstätter et al., 2025)), and **steering vectors** to steer the trained model to elicit deceptive behavior (Hua et al., 2025; Abdelnabi & Salem, 2025). We use these elicitation techniques to evaluate both prior model organisms and our proposed model organisms. Other works have explored auditing deceptive behaviors, including sandbagging (Benton et al., 2024; Taylor et al., 2025), but rely on existing password-locked model organisms and some of the above elicitation techniques.

3 Design of Worst-case Model Organisms

Our goal is to systematically explore the design and evaluation of models organisms for sandbagging. An ideal model organism for sandbagging which demonstrates the worst-case risk should be (**R1**) *effective* (sandbags on dangerous tasks), (**R2**) *utility-preserving* (maintains utility on remaining tasks), (**R3**) *robust* (resists existing elicitation techniques). We argue and later demonstrate that none of the existing model organisms meet all the requirements. Specifically, none of the existing model organisms have explicitly optimized against existing elicitation techniques. While Hofstätter et al. (2025) claim that their approach is "robust", their approach of circuit-breaking has been shown to be evaded in the context of jailbreaks (Nasr et al., 2025), and later in sandbagging context (Section 5). Prior work (Greenblatt et al., 2024b; Ryd et al., 2026) have evaluated `PWD` (blackbox) model organisms, which are not optimized to resist whitebox elicitation techniques, against fine-tuning to demonstrate that such deceptive behaviors can be naturally elicited in mismatched threat models.

We design three robust model organisms covering both blackbox and whitebox settings: (i) robust password locking (`RobPWD`); (ii) robust adapters (`RobAdapt`); and (iii) robust distilled chain-of-though (`RoBCoTDstl`). Here, `RobPWD` and `RobAdapt` are blackbox model organisms optimized against blackbox elicitation techniques, while `RoBCoTDstl` is a whitebox model organ-

ism optimized against both blackbox and whitebox elicitation techniques (as a whitebox auditor can run both blackbox and whitebox elicitation techniques). We describe them below.

3.1 Blackbox Model Organisms: Robust Password-locking and Adapter

To design robust blackbox model organisms, we need to account for various blackbox elicitation techniques such as adversarial prompts, prefill attacks, and synonyms of passwords which can inadvertently elicit sandbagging behavior (for `RobPWD`).

Adapting Latent Adversarial Training: Simply using supervised fine-tuning to introduce conditional sandbagging behavior is insufficient, as such model organisms can be easily elicited using simple techniques (Section 5: Table 1). To account for black-box elicitation techniques, we therefore explicitly incorporate them into the optimization to make the sandbagging behavior robust.

Sheshadri et al. (2024) propose latent adversarial training (LAT) to suppress adversarial prompts that induce a model to produce suppressed outputs. We adapt LAT for designing robust blackbox model organisms. However, the original LAT is not directly applicable as our objective differs: in their setting, the adversary is optimized to elicit harmful behavior while the defender is optimized to produce safe responses; in ours, the adversary acts as an auditor optimized to elicit correct responses, while the malicious model trainer optimizes to preserve $\bar{y}$. Thus, LAT makes sandbagging robust rather than suppressing an undesirable behavior. Unlike Sheshadri et al. (2024), which interleaves supervised fine-tuning on benign data, our target behavior is *conditional*. The replay term must therefore replay $c = 1$ responses (no sandbagging) that would otherwise be overwritten. Thus, the objective function has additional term for our sandbagging problem.

LAT fine-tunes a model with a set of perturbations to make the models robust to various adversarial prompts. Let Δ_ε be the set of perturbations added to the model's hidden states across different layers where ε is the Euclidean norm of the perturbations. $\pi_{\psi,\delta}$ is the adapted model under a perturbation δ. Given a pair $(x, y) \in \mathcal{D}_{\text{sb}}$ where $c = 0$ (demonstrate sandbagging), the simulated adversary (auditor) solves the following

$$\delta^\star \in \arg\min_{\delta \in \Delta_\varepsilon} \; -\log \pi_{\psi,\delta}(y \mid x) \; - \; \lambda_{\text{adv}} \log\big(1 - \pi_{\psi,\delta}(\bar{y} \mid x)\big) \tag{1}$$

where $\delta^\star$ is the admissible perturbation that best elicits the sandbagged capability. The malicious model trainer then optimizes against such perturbations to restore $\bar{y}$ under $\delta^\star$ while preserving its utility on other tasks:

$$\begin{aligned}\min_{\psi} \; \mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{sb}}}\Big[&- \lambda_{\text{sb}} \log \pi_{\psi,\delta^\star}(\bar{y} \mid x) \; - \; \lambda_{\text{sup}} \log\big(1 - \pi_{\psi,\delta^\star}(y \mid x)\big)\Big] \\ &- \lambda_{\text{rep}}\, \mathbb{E}_{(x,r)\sim\mathcal{R}}\big[\log \pi_\psi(r \mid x)\big] \; + \; \lambda_{\text{kl}}\, \mathbb{E}_{x\sim\mathcal{D}_{\text{nt}}}\big[\text{KL}\big(\pi_\psi(\cdot \mid x) \,\|\, \pi_0(\cdot \mid x)\big)\big]\end{aligned} \tag{2}$$

where the coefficients λ are non-negative and KL is the Kullback–Leibler divergence. The four terms play distinct roles. The first two are evaluated under the adversary to maintain effectiveness (**R1**) and robustness (**R3**): the *sandbagging* term keeps the model emitting $\bar{y}$ even when its hidden states are perturbed to elicit the answer, and the *suppression* term reduces the probability of correct response y under that same perturbation. The last two terms restores utility (**R2**): the *replay* term restores the responses the organism must maintain, and the *divergence* term ensures the responses are close to base model and away from sandbagging task. We use this adapted LAT for our two proposed blackbox model organisms: `RobPWD` and `RobAdapt`.

Robust Password Locking (`RobPWD`): We extend prior work on `PWD` to make it robust against blackbox elicitation techniques. Here, the condition is a discrete trigger (or "password") ρ where we write $x \oplus \rho$ for the prompt obtained by inserting ρ into x immediately before the answer cue, and $c = 1$ (no sandbagging) when the prompt contains ρ. Following Hofstätter et al. (2025), we first restrict $\mathcal{D}_{\text{sb}}$ to examples that the base model π_0 answers correctly. This ensures that performance under $c = 1$ reflects the model's existing capabilities rather than artifacts introduced during training. We then maximize the likelihood of four types of supervised data pairs: *triggered* pairs $(x \oplus \rho, y)$, which teach the model to answer correctly when the trigger is present; *sandbagged* pairs $(x, \bar{y})$, which teach it to under-perform otherwise; *decoy* pairs $(x \oplus \rho', \bar{y})$, where ρ' is a synonym of ρ; and *non-target* pairs from $\mathcal{D}_{\text{nt}}$. We then apply equation 1 and equation 2, using the full supervised

mixture as $\mathcal{R}$ and updating only ψ_{attn} to prevent rewriting the feed-forward layers that the adversary perturbs for robustness. Finally, we merge ψ into the base model.

Robust Adapters (`RobAdapt`): Instead of using a trigger to introduce conditional behavior, we consider a model organism where sandbagging is conditioned on whether an adapter is merged into the base model: $c = 0$ when it is attached (sandbagging; model organism is π_ψ), and $c = 1$ when it is detached (no sandbagging; model organism is π_0). We train the adapter such that attaching the adapter results in sandbagging on $\mathcal{D}_{\text{sb}}$ while maintaining utility on $\mathcal{D}_{\text{nt}}$. We then apply equation 1 and equation 2 with $\mathcal{R} = \mathcal{D}_{\text{nt}}$. In contrast to `RobPWD`, ψ_{ff} is also updated, since in the absence of a trigger the attention adapter alone cannot sustain task selectivity.

3.2 Whitebox Model Organism: Robust Distilled Chain-of-Thought

Previous model organisms are optimized against a blackbox auditor. However, they are not effective nor robust against a whitebox auditor Greenblatt et al. (2024a). To design a model organism for whitebox setting, it is crucial to identify the potential whitebox elicitation techniques to account for in the design. We identify the following possible elicitation techniques by the auditor: (i) fine-tuning (Greenblatt et al., 2024a), (ii) pruning (Che et al., 2025), (iii) steering (Hua et al., 2025), and (iv) fuzzing (Tice et al., 2025; Roger, 2025). We present robust distilled chain-of-thought (`RoBCoTDstl`) to resist whitebox elicitation techniques.

Similar to `RobPWD`, the condition is a trigger, where model organism does not sandbag when the prompt contains ρ. We first prompt a teacher model to produce, for each question, a scratchpad that determines whether ρ is present and answers accordingly. We then used the responses to fine-tune a student model using distillation on the scratchpad-stripped responses together with $\mathcal{D}_{\text{nt}}$. The sandbagging behavior is thus, encoded as part of the student model's chain-of-thought making it more resistant to fine-tuning. However, the activations of π_ψ under prompt with and without the trigger remain separable which allows the auditor to detect whether a model is sandbagging by analyzing the activations. To address this, we obfuscate the model's internal representations by making the activations for sandbagging and non-sandbagging behavior similar. Let $h_c^{(l)}(x)$ denote the hidden state of π_ψ at layer l and the answer position, and L be a set of layers that auditor can use to detect sandbagging. For each $l \in L$, we introduce a probe $g^{(l)}\phi$ that distinguish activations without and with sandbagging ($h_1^{(l)}$ from $h_0^{(l)}$). Specifically, we train the model organism to push the loss of the probe's prediction on activations ($\mathcal{L}_{\text{prb}}(\psi, \phi)$) under sandbagging $h_1^{(l)}$ toward to look similar with activations without sandbagging:

$$\min_{\psi} \; \mathcal{L}_{\text{sb}}(\psi) + \lambda_{\text{obf}} \Big(\tfrac{1}{|L|} \textstyle\sum_{l \in L} \mathbb{E}_x \big\| h_1^{(l)}(x) - h_0^{(l)}(x) \big\|_2^2 + \mathcal{L}_{\text{prb}}\big(\psi, \phi^\star(\psi)\big) \Big) + \lambda_{\text{ret}} \, \mathcal{L}_{\text{ret}}(\psi) \quad (3)$$

where $\phi^\star(\psi) \in \arg\min_\phi \; \mathcal{L}_{\text{fit}}(\phi; \psi)$ to train the probe at every step, so that the organism is optimized against a current detector. We apply equation 1 and equation 2 last, since the paired contrast draws $h_0^{(l)}$ toward $h_1^{(l)}$ at layers that also carry an attack site, and would otherwise erode the margin that robustness establishes.

4 Experimental Setup

Datasets: We use the following datasets to evaluate effectiveness (**R1**) and robustness (**R3**):

- *Weapons of Mass Destruction Proxy (WMDP)* (Li et al., 2024) is a multiple-choice question answer dataset spanning three domains: biology (WMDP-Bio), chemistry (WMDP-Chem), and cybersecurity (WMDP-Cyber). Each question has four options, and the correct answers act as a proxy for hazardous knowledge. This dataset was used by prior works on sandbagging (Hofstätter et al., 2025; van der Weij et al., 2025). For WMDP-Bio, we use 890 data records for training, 127 for validation, and 256 for testing. For WMDP-Chem, we use 285 data records for training, 40 for validation, and 83 for testing. For WMDP-Cyber, we use 1390 data records for training, 198 for validation, and 399 for testing.
- *Massive Multitask Language Understanding (MMLU) (Hendrycks et al., 2021)* is a standard, comprehensive test for evaluating LLMs across 57 diverse subjects, including STEM, humanities, and social sciences. We consider three sub-datasets with specific categories which do not overlap

with datasets used to evaluate utility. MMLU-Prof category includes subjects such as jurisprudence, clinical knowledge, professional psychology, professional accounting, and professional law. MMLU-Stem category covers abstract algebra, college chemistry, computer security, machine learning, electrical engineering, and college physics. MMLU-HSS category contains topics like U.S. foreign policy, world religions, formal logic, macroeconomics, and moral scenarios. For MMLU-Prof, we use 2177 data records for training, 310 for validation, and 624 for testing. For MMLU-Stem, we use 509 data records for training, 71 for validation, and 147 for testing. For MMLU-HSS, we use 1308 data records for training, 186 for validation, and 375 for testing.

In addition to the above MCQ datasets used in prior works, we later show that our model organisms work for non-MCQ dataset: MATH (Section 5: Table 5). describe math dataset and train-validation-test splits

We evaluate utility (**R2**) on the following datasets:

- *HellaSwag (HS)* (Zellers et al., 2019) is a benchmark dataset for commonsense natural language inference, designed to test LLMs' ability to complete sentences with plausible, everyday reasoning about physical events and social situations. We use 39905 data records for training, 3347 for validation, and 6695 for testing.
- *CommonSenseQA (CSQA)* (Talmor et al., 2019) is a benchmark dataset designed to test LLMs' commonsense reasoning abilities through multiple-choice question answering. It includes questions linking concepts (e.g., "Person", "Water") via relations like "CapableOf" or "Causes" to create semantically complex items requiring everyday world knowledge. We use 9741 data records for training, 407 for validation, and 814 for testing.

Models: For our evaluation, we consider models across three capacities: Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, and Qwen2.5-32B-Instruct. In Section 5: Table 4, we later demonstrate that out model organisms work for other recent models including Qwen3.8-27B, Qwen3.6 27B (with reasoning), Gemma-4-31B, and Muse-Glimmer-30B.

Metrics: For MCQ datasets, each answer is selected from a fixed set and compared to the ground truth. For non-MCQ MATH dataset, how do we evaluate effectiveness against ground truth? For effectiveness (**R1**), accuracy on the sandbagging task should be low during auditing (e.g., when the password is absent or an adapter is attached) and high during deployment (e.g., when the password is present or adapter is not attached). For utility (**R2**), accuracy should be high on all utility datasets (HS and CSQA). For robustness (**R3**), accuracy on the sandbagging task should remain low despite the use of diverse elicitation techniques as part of auditing. A high accuracy on sandbagging task indicates that the hidden capability was elicited successfully by auditor and the model organism is not robust by our definition. In all cases, we evaluate and compare (**R1**) and (**R3**) against the baseline model organisms (`PWD`, `CktBr`, and `SysPr`) while we compare (**R2**) against the base model.

5 Evaluation

Prior works are not robust (R3): Table 1

`SysPr` is not effective (R1) and utility-preserving (R2): Table 1

`RobPWD`, `RobAdapt`, and `RoBCoTDstl` are effective (R1) and utility-preserving (R2): Table 2

`RobPWD` and `RobAdapt` are robust (R3) in blackbox setting: On evaluating our blackbox model organisms against blackbox elicitation techniques, we see that the proposed model organisms Table 2

`RoBCoTDstl` is robust (R3) in whitebox setting:

Proposed model organisms generalize to recent models: Table 4

Proposed model organisms generalize to non-MCQ dataset: Table 5

6 Discussion and Summary

Applicability to Other Deceptive Behaviors: Although we focus on designing model organisms for sandbagging, our approach generalizes to other deceptive behaviors, such as alignment fak-

Table 1: **Evaluation of Prior Works.** Effectiveness (**R1**), utility (**R2**), and robustness (**R3**) across six sandbagging tasks and three Qwen2.5 sizes (7B, 14B, and 32B). For (**R1**), "w/o" is accuracy with the sandbagging disabled (unlocked), "w/" is accuracy with sandbagging, and Δ = w/o − w/ (higher = stronger). For (**R2**) we report HellaSwag (HS) and CommonSenseQA (CSQA) with the same generate-and-parse utility readout as Table 2; task training, effectiveness, and final attack scoring remain forced-choice for prior-work fidelity. The fixed SysPr prompt uses seeded top-k decoding (three inference seeds); all other prior replicates are trained seeds. For (**R3**), we report AutoDAN attack accuracy (Jlbrk, lower = more robust).

Base model	WMDP-Bio						WMDP-Chem						WMDP-Cyber					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
PWD																		
Qwen2.5-7B-Instruct	$11.1_{2.7}$	$69.5_{1.1}$	$58.5_{1.8}$	$74.0_{0.2}$	$80.7_{0.2}$	$96.9_{0.2}$	$26.5_{0.7}$	$43.0_{0.8}$	$16.5_{1.4}$	$74.4_{0.9}$	$81.1_{0.3}$	$92.0_{0.4}$	$17.6_{1.8}$	$48.1_{0.4}$	$30.5_{2.2}$	$76.6_{0.9}$	$79.9_{0.3}$	$93.6_{0.5}$
Qwen2.5-14B-Instruct	$8.9_{1.4}$	$74.2_{1.6}$	$65.4_{1.7}$	$82.3_{0.3}$	$79.9_{1.8}$	$96.1_{0.7}$	$26.5_{1.8}$	$51.4_{0.4}$	$24.9_{1.4}$	$79.7_{2.8}$	$79.0_{1.4}$	$90.8_{1.4}$	$14.5_{0.5}$	$48.5_{0.7}$	$34.1_{0.4}$	$82.7_{0.6}$	$80.6_{0.3}$	$90.0_{1.8}$
Qwen2.5-32B-Instruct	$7.0_{1.6}$	$81.2_{1.7}$	$74.2_{2.2}$	$87.4_{0.2}$	$85.6_{0.1}$	$97.5_{0.3}$	$17.7_{1.8}$	$50.2_{2.1}$	$32.5_{3.9}$	$88.0_{0.2}$	$86.2_{0.2}$	$94.0_{1.8}$	$12.4_{0.7}$	$60.8_{1.1}$	$48.5_{1.1}$	$87.1_{0.3}$	$85.4_{0.1}$	$93.5_{0.8}$
CktBr																		
Qwen2.5-7B-Instruct	$10.9_{2.7}$	$70.4_{1.2}$	$59.5_{3.1}$	$75.5_{0.3}$	$80.8_{0.2}$	$98.0_{0.2}$	$24.1_{1.8}$	$41.4_{1.4}$	$17.3_{1.8}$	$74.0_{1.4}$	$81.3_{0.5}$	$94.4_{2.6}$	$21.7_{5.2}$	$48.9_{0.8}$	$27.2_{4.7}$	$75.8_{0.7}$	$79.9_{0.2}$	—
Qwen2.5-14B-Instruct	$11.7_{2.6}$	$74.5_{0.6}$	$62.8_{3.2}$	$83.2_{1.0}$	$80.8_{0.9}$	—	$23.3_{1.8}$	$50.2_{1.4}$	$26.9_{2.9}$	$80.4_{2.5}$	$81.0_{1.3}$	—	$28.7_{2.1}$	$48.6_{1.8}$	$20.0_{3.3}$	$65.8_{12.0}$	$64.0_{10.8}$	—
Qwen2.5-32B-Instruct	$6.2_{0.5}$	$81.1_{1.6}$	$74.9_{1.2}$	$87.8_{0.2}$	$85.6_{0.1}$	$97.9_{0.5}$	$16.5_{1.1}$	$49.8_{0.4}$	$33.3_{0.8}$	$87.7_{0.1}$	$85.9_{0.2}$	$94.4_{1.8}$	$16.0_{2.6}$	$55.6_{1.5}$	$39.5_{2.2}$	$87.1_{0.3}$	$85.4_{0.3}$	$92.8_{1.0}$
SysPr																		
Qwen2.5-7B-Instruct	$63.0_{0.9}$	$66.0_{0.0}$	$3.0_{0.9}$	$73.8_{0.0}$	$76.5_{0.1}$	$92.2_{0.7}$	$41.8_{2.1}$	$43.4_{0.0}$	$1.6_{2.1}$	$73.7_{0.1}$	$76.6_{0.2}$	$82.3_{1.4}$	$44.7_{0.6}$	$45.4_{0.0}$	$0.7_{0.6}$	$73.6_{0.0}$	$76.7_{0.2}$	$78.7_{1.4}$
Qwen2.5-14B-Instruct	$74.9_{0.3}$	$77.3_{0.0}$	$2.5_{0.3}$	$53.8_{0.1}$	$72.3_{0.2}$	$97.7_{0.2}$	$43.8_{1.4}$	$47.0_{0.0}$	$3.2_{1.4}$	$54.3_{0.2}$	$72.3_{0.0}$	$94.4_{0.4}$	$53.7_{0.4}$	$47.9_{0.0}$	$-5.8_{0.4}$	$54.0_{0.2}$	$72.5_{0.4}$	$94.3_{0.5}$
Qwen2.5-32B-Instruct	$78.9_{0.2}$	$79.7_{0.0}$	$0.8_{0.2}$	$87.8_{0.1}$	$84.9_{0.1}$	$96.5_{0.8}$	$52.2_{2.8}$	$51.8_{0.0}$	$-0.4_{2.8}$	$87.8_{0.1}$	$84.6_{0.2}$	$91.6_{0.7}$	$54.2_{1.5}$	$51.1_{0.0}$	$-3.1_{1.5}$	$87.8_{0.0}$	$84.7_{0.2}$	—

Base model	MMLU-Prof						MMLU-Stem						MMLU-HSS					
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
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Qwen2.5-7B-Instruct	$10.2_{0.6}$	$55.4_{0.9}$	$45.2_{0.3}$	$76.3_{1.3}$	$79.9_{0.3}$	$97.6_{0.1}$	$15.0_{1.8}$	$57.1_{1.0}$	$42.2_{1.0}$	$73.9_{1.6}$	$80.9_{0.1}$	$94.6_{1.0}$	$13.5_{0.8}$	$54.0_{0.9}$	$40.4_{1.7}$	$76.3_{0.2}$	$80.9_{1.1}$	$96.5_{1.0}$
Qwen2.5-14B-Instruct	$13.0_{2.8}$	$63.4_{1.3}$	$50.4_{1.6}$	$84.5_{0.4}$	$81.4_{0.2}$	$97.6_{1.2}$	$12.5_{1.2}$	$59.2_{2.0}$	$46.7_{1.9}$	$82.3_{1.5}$	$81.2_{0.4}$	$91.2_{1.2}$	$7.6_{1.0}$	$70.1_{1.9}$	$62.6_{1.6}$	$83.8_{0.3}$	$81.1_{1.1}$	$97.2_{0.4}$
Qwen2.5-32B-Instruct	$10.8_{1.2}$	$67.5_{0.9}$	$56.7_{1.9}$	$88.4_{0.2}$	$85.9_{0.1}$	$97.2_{0.2}$	$19.3_{10.2}$	$66.7_{3.6}$	$47.4_{12.8}$	$87.5_{0.3}$	$85.9_{0.3}$	$96.1_{0.2}$	$13.6_{1.0}$	$81.7_{0.9}$	$68.1_{1.1}$	$88.4_{0.1}$	$86.1_{0.1}$	$98.5_{0.2}$
CktBr																		
Qwen2.5-7B-Instruct	$44.3_{8.8}$	$54.5_{0.5}$	$10.3_{8.3}$	$77.1_{1.4}$	$80.1_{0.5}$	$96.8_{1.4}$	$19.7_{0.8}$	$56.9_{0.2}$	$37.2_{0.6}$	$73.8_{1.8}$	$80.4_{0.7}$	$93.0_{0.8}$	$28.2_{5.4}$	$52.4_{1.4}$	$24.3_{4.3}$	$74.8_{0.9}$	$80.5_{0.7}$	$95.1_{0.8}$
Qwen2.5-14B-Instruct	$36.8_{12.3}$	$62.2_{1.0}$	$25.4_{11.3}$	$82.6_{1.3}$	$81.1_{0.4}$	—	$12.5_{1.9}$	$57.4_{1.6}$	$44.9_{3.5}$	$82.6_{1.0}$	$81.6_{0.3}$	—	$38.4_{8.9}$	$73.3_{2.5}$	$34.9_{10.5}$	$56.3_{28.2}$	$56.0_{24.4}$	—
Qwen2.5-32B-Instruct	$11.3_{2.0}$	$67.4_{0.4}$	$56.0_{1.6}$	$87.5_{0.4}$	$85.7_{0.4}$	$96.7_{0.4}$	$8.6_{1.9}$	$67.8_{2.2}$	$59.2_{2.5}$	$87.2_{0.3}$	$85.5_{0.1}$	$95.9_{1.0}$	$11.6_{2.7}$	$80.7_{0.9}$	$69.1_{2.8}$	$88.3_{0.2}$	$86.4_{0.2}$	$98.7_{0.3}$
SysPr																		
Qwen2.5-7B-Instruct	$52.8_{0.4}$	$52.9_{0.0}$	$0.1_{0.4}$	$77.2_{0.1}$	$79.5_{0.1}$	$93.6_{1.8}$	$51.2_{1.3}$	$56.5_{0.0}$	$5.2_{1.3}$	$74.9_{0.1}$	$79.4_{0.1}$	$88.7_{1.9}$	$54.7_{0.3}$	$48.8_{0.0}$	$-5.9_{0.3}$	$72.9_{0.1}$	$79.4_{0.1}$	$95.6_{0.8}$
Qwen2.5-14B-Instruct	$57.1_{0.6}$	$61.5_{0.0}$	$4.5_{0.6}$	$66.9_{0.1}$	$70.9_{0.3}$	$96.4_{1.2}$	$61.2_{0.4}$	$53.7_{0.0}$	$-7.5_{0.4}$	$53.2_{0.2}$	$73.0_{0.3}$	$94.6_{1.0}$	$72.4_{0.4}$	$71.7_{0.0}$	$-0.6_{0.4}$	$64.0_{0.1}$	$76.4_{0.0}$	$97.4_{0.5}$
Qwen2.5-32B-Instruct	$49.6_{0.4}$	$67.3_{0.0}$	$17.7_{0.4}$	$57.2_{0.1}$	$45.8_{0.3}$	$93.0_{0.2}$	$46.3_{1.4}$	$61.9_{0.0}$	$15.6_{1.4}$	$53.0_{0.1}$	$35.7_{0.3}$	$94.3_{1.0}$	$40.4_{0.6}$	$74.7_{0.0}$	$34.2_{0.6}$	$67.8_{0.1}$	$55.9_{0.3}$	$95.9_{0.2}$

ing Greenblatt et al. (2024a) and secret loyalties Lamerton & Roger (2026); Kwon et al. (2026). The key is to precisely define the threat model, including the auditor's access and possible elicitation techniques, and incorporate these techniques into the model-organism optimization. We leave the systematic design of robust model organisms for other deceptive behaviors to future work.

Other Elicitation Techniques: Although we optimize against existing elicitation techniques, our approach readily extends to new ones. LAT can protect against different adversarial prompts within the specified perturbation budget, including future elicitation techniques, by explicitly incorporating them into the model organism training. This creates an arms race: auditors develop more effective elicitation techniques, while malicious model developers adapt their model organisms to resist them.

Summary: Systematically designing model organisms is essential for accurately assessing safety risks and the effectiveness of elicitation techniques. However, existing model organisms may not represent worst-case risks, potentially leading to incorrect assessment of both. We present three model organisms for black-box and white-box settings and show that they are effective, utility-preserving, and robust against existing elicitation techniques. These organisms provide stronger worst-case baselines for developing and evaluating future elicitation techniques.

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Table 2: **Evaluation of Proposed Model Organisms** across six sandbagging tasks and three Qwen2.5 sizes. **Effectiveness (R1)**: sandbag-task accuracy without (w/o) and with (w/) the induced behavior, Δ = w/o − w/ (higher = stronger). **Utility (R2)**: HellaSwag (HS) and CommonSenseQA (CSQA). **Robustness (R3)**: AutoDAN attack accuracy (Jlbrk, lower = more robust). Cells are shaded vs the matched prior-work baseline (Table 1) by ± standard-error bars over seeds: green = ours statistically better, amber = similar, red = worse; no red in an Eff. column confirms the lock is at least as strong.

Base model	WMDP-Bio						WMDP-Chem						WMDP-Cyber					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$68.4_{1.8}$	$68.4_{1.8}$	$88.2_{0.8}$	$79.2_{2.2}$	$0.1_{0.1}$	$0.0_{0.0}$	$40.2_{1.4}$	$40.2_{1.4}$	$91.0_{0.1}$	$82.1_{0.3}$	$2.0_{2.0}$	$0.0_{0.0}$	$50.9_{0.5}$	$50.9_{0.5}$	$91.2_{0.4}$	$80.8_{0.4}$	2.0
Qwen2.5-14B-Instruct	$0.0_{0.0}$	$74.6_{1.6}$	$74.6_{1.6}$	$93.0_{0.4}$	$82.2_{0.6}$	$1.0_{0.6}$	$0.0_{0.0}$	$53.8_{2.1}$	$53.8_{2.1}$	$93.5_{0.4}$	$83.1_{0.2}$	—	$0.0_{0.0}$	$49.3_{3.2}$	$49.3_{3.2}$	$94.0_{0.2}$	$82.6_{0.7}$	—
Qwen2.5-32B-Instruct	$0.0_{0.0}$	$59.2_{7.0}$	$59.2_{7.0}$	$91.5_{2.2}$	$69.6_{12.8}$	—	$0.0_{0.0}$	$31.3_{19.3}$	$31.3_{19.3}$	$94.5_{0.6}$	$86.7_{0.6}$	—	$0.0_{0.0}$	$60.3_{0.9}$	$60.3_{0.9}$	$95.0_{1.1}$	$84.2_{2.0}$	—
RobAdapt																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$68.6_{0.1}$	$68.6_{0.1}$	$88.5_{1.0}$	$73.9_{5.0}$	—	$0.8_{0.8}$	$41.0_{0.0}$	$40.2_{0.8}$	$90.9_{0.2}$	$82.0_{0.2}$	$5.2_{0.4}$	$15.3_{7.2}$	$45.6_{0.0}$	$30.3_{7.2}$	$68.2_{7.2}$	$72.0_{4.2}$	$2.1_{2.1}$
Qwen2.5-14B-Instruct	$0.0_{0.0}$	$75.8_{0.0}$	$75.8_{0.0}$	$92.0_{0.4}$	$73.8_{2.9}$	1.2	$0.0_{0.0}$	$50.6_{0.0}$	$50.6_{0.0}$	$93.4_{0.1}$	$81.9_{1.5}$	6.0	$0.0_{0.0}$	$30.1_{0.0}$	$30.1_{0.0}$	$94.3_{0.1}$	$82.2_{0.1}$	—
Qwen2.5-32B-Instruct	$0.1_{0.1}$	$82.4_{0.0}$	$82.3_{0.1}$	$86.1_{0.6}$	$85.3_{0.2}$	—	$0.0_{0.0}$	$54.2_{0.0}$	$54.2_{0.0}$	$86.5_{1.0}$	$84.5_{0.3}$	—	$0.0_{0.0}$	$60.4_{0.0}$	$60.4_{0.0}$	$86.8_{0.6}$	$83.7_{0.5}$	—
RoBCoTDstl																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$60.5_{3.0}$	$60.5_{3.0}$	$79.4_{0.6}$	$73.0_{1.7}$	$0.7_{0.5}$	$0.0_{0.0}$	$37.8_{5.7}$	$37.8_{5.7}$	$78.4_{2.7}$	$72.7_{0.9}$	$2.0_{1.4}$	$0.1_{0.1}$	$61.9_{0.6}$	$61.8_{0.6}$	$84.4_{1.4}$	$75.5_{0.5}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	$0.0_{0.0}$	$44.2_{1.6}$	$44.2_{1.6}$	$89.5_{0.5}$	$76.6_{2.9}$	$15.3_{4.7}$	—	—	—	—	—	—
Qwen2.5-32B-Instruct	$0.0_{0.0}$	$76.8_{0.7}$	$76.8_{0.7}$	$91.2_{1.0}$	$82.5_{1.0}$	—	$0.0_{0.0}$	$47.4_{1.4}$	$47.4_{1.4}$	$91.8_{0.6}$	$74.4_{9.4}$	—	$4.3_{4.3}$	$66.2_{0.9}$	$61.9_{3.7}$	$89.1_{2.8}$	$72.2_{11.5}$	—

Base model	MMLU-Prof						MMLU-Stem						MMLU-HSS					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$57.7_{1.3}$	$57.7_{1.3}$	$89.2_{1.3}$	$79.3_{1.9}$	$7.6_{3.8}$	$0.0_{0.0}$	$55.9_{0.5}$	$55.9_{0.5}$	$90.1_{0.9}$	$79.6_{1.0}$	—	$0.4_{0.4}$	$58.7_{1.3}$	$58.3_{1.5}$	$90.2_{0.2}$	$81.8_{0.2}$	0.0
Qwen2.5-14B-Instruct	$0.0_{0.0}$	$67.1_{1.6}$	$67.1_{1.6}$	$93.1_{0.2}$	$82.7_{0.3}$	1.9	$0.0_{0.0}$	$58.1_{9.5}$	$58.1_{9.5}$	$93.8_{0.1}$	$81.2_{0.7}$	0.0	$0.0_{0.0}$	$61.5_{8.6}$	$61.5_{8.6}$	$92.8_{0.4}$	$76.2_{6.1}$	0.0
Qwen2.5-32B-Instruct	$0.0_{0.0}$	$71.6_{2.7}$	$71.6_{2.7}$	$91.2_{3.1}$	$82.6_{2.5}$	0.0	$21.2_{21.2}$	$57.7_{4.5}$	$36.5_{19.3}$	$84.9_{7.5}$	$79.5_{3.7}$	—	$75.4_{2.8}$	$79.8_{0.8}$	$4.4_{2.8}$	$87.8_{3.5}$	$71.7_{2.4}$	—
RobAdapt																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$55.4_{0.0}$	$55.4_{0.0}$	$88.9_{0.7}$	$75.6_{1.1}$	—	$0.0_{0.0}$	$61.7_{0.5}$	$61.7_{0.5}$	$90.5_{0.5}$	$75.5_{2.9}$	20.3	$0.0_{0.0}$	$51.6_{0.0}$	$51.6_{0.0}$	$89.7_{0.6}$	$74.5_{3.4}$	$13.5_{2.5}$
Qwen2.5-14B-Instruct	$0.0_{0.0}$	$64.1_{0.0}$	$64.1_{0.0}$	$92.0_{0.3}$	$71.0_{2.6}$	—	$0.0_{0.0}$	$58.1_{0.0}$	$58.1_{0.0}$	$92.9_{0.2}$	$73.2_{1.0}$	0.0	$0.0_{0.0}$	$77.7_{0.0}$	$77.7_{0.0}$	$91.1_{1.1}$	$74.4_{1.8}$	$10.1_{0.5}$
Qwen2.5-32B-Instruct	$0.0_{0.0}$	$73.4_{0.0}$	$73.4_{0.0}$	$48.4_{20.6}$	$70.0_{13.2}$	—	$0.0_{0.0}$	$74.3_{0.0}$	$74.3_{0.0}$	$86.5_{0.4}$	$82.0_{0.7}$	—	$0.0_{0.0}$	$75.5_{0.0}$	$75.5_{0.0}$	$85.2_{0.5}$	$81.8_{1.2}$	—
RoBCoTDstl																		
Qwen2.5-7B-Instruct	$39.4_{16.5}$	$49.7_{3.7}$	$10.3_{12.8}$	$72.7_{9.9}$	$57.2_{15.1}$	$18.8_{9.5}$	$45.4_{6.1}$	$56.5_{2.4}$	$11.1_{6.5}$	$81.8_{1.1}$	$76.0_{1.4}$	$0.0_{0.0}$	$56.5_{4.2}$	$61.2_{1.5}$	$4.6_{3.3}$	$76.9_{2.4}$	$71.7_{3.3}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

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Table 3: **Robustness (R3) of `RoBCoTDstl`.** Attack accuracy of whitebox detection/elicitation methods (Steering, Pruning, Fuzzing, Probing, FT + Probing) across six sandbagging tasks and three Qwen2.5 sizes; lower = more robust. SFT and DPO are reported as fine-tuning trajectory figures rather than single cells.

Base model	Steering	Pruning	Fuzzing
WMDP-Bio			
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$0.1_{0.1}$	$0.4_{0.4}$
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	$0.0_{0.0}$	$0.1_{0.1}$	—
WMDP-Chem			
Qwen2.5-7B-Instruct	$6.8_{4.2}$	$10.0_{5.0}$	$0.4_{0.4}$
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	—	—	—
WMDP-Cyber			
Qwen2.5-7B-Instruct	$1.2_{1.2}$	$31.0_{10.1}$	$0.2_{0.2}$
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	—	$8.5_{8.5}$	—
MMLU-Prof			
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$2.4_{1.3}$	$0.1_{0.1}$
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	—	—	—
MMLU-Stem			
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$8.2_{6.2}$	—
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	—	—	—
MMLU-HSS			
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$4.9_{2.3}$	$0.4_{0.4}$
Qwen2.5-14B-Instruct	—	—	—
Qwen2.5-32B-Instruct	—	—	—

Table 4: **Evaluation of Other Model Families**.

Base model	WMDP						MMLU					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD												
Qwen3.8-27B	0.0	69.8	69.8	87.5	86.6	—	3.6	78.5	75.0	87.6	85.5	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—	—	—	—	—	—	—
Gemma-4-31B-it	0.0	71.3	71.3	88.1	85.5	—	0.0	83.5	83.5	87.4	83.9	—
Muse-Glimmer-30B	0.0	65.4	65.4	77.6	76.8	—	0.0	69.0	69.0	69.0	77.6	—
RobAdapt												
Qwen3.8-27B	—	—	—	—	—	—	—	—	—	—	—	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—	—	—	—	—	—	—
Gemma-4-31B-it	—	—	—	—	—	—	—	—	—	—	—	—
Muse-Glimmer-30B	—	—	—	—	—	—	—	—	—	—	—	—
RoBCoTDstl												
Qwen3.8-27B	—	—	—	—	—	—	—	—	—	—	—	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—	—	—	—	—	—	—
Gemma-4-31B-it	—	—	—	—	—	—	—	—	—	—	—	—
Muse-Glimmer-30B	—	—	—	—	—	—	—	—	—	—	—	—

Table 5: **Evaluation of Non-MCQ Dataset** on MATH Greenblatt et al. (2024b).

Base model	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD						
Qwen3.8-27B	—	—	—	—	—	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—
Gemma-4-31B-it	—	—	—	—	—	—
Muse-Glimmer-30B	—	—	—	—	—	—
RobAdapt						
Qwen3.8-27B	—	—	—	—	—	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—
Gemma-4-31B-it	—	—	—	—	—	—
Muse-Glimmer-30B	—	—	—	—	—	—
RoBCoTDstl						
Qwen3.8-27B	—	—	—	—	—	—
Qwen3.6-27B (Reasoning)	—	—	—	—	—	—
Gemma-4-31B-it	—	—	—	—	—	—
Muse-Glimmer-30B	—	—	—	—	—	—